

Corrections to the Thesis: Fractional Gaussian Fields on the Vicsek Fractal 11th June 2025

Bigger corrections:

Lemma 2.14: Let $v: V^* \rightarrow \mathbb{R}$ be a harmonic function on the Vicsek set. If $v(x) = v(y_i) = 0$ for all $y_i \in V_{m,x}$ then $v(\omega) = 0$ for all $\omega \in \overline{T_x}$ where $T_x = \left(\bigcup_{k>0} S^k(x)\right) \cup \left(\bigcup_i \{y_i\}\right)$.

Proof. It is clear that $v(\omega) = 0$ for any $\omega \in T_x$, since the value of the points $S^k(x)$ with $k > 0$ depends on the value of x and its neighbours.

So we need to check the points in $\overline{T_x} \setminus T_x$. Now since T_x is dense in $\overline{T_x}$, then for any $\omega \in \overline{T_x}$ we have ... □

This should be corrected to:

If $x, y \in K$ then $\{x\} = x_{i_1, i_2, \dots}$ and $\{y\} = y_{j_1, j_2, \dots}$. Furthermore, if $x \sim_m y$ they must share an α such $i_m = j_m$ for $m \leq \alpha$.

Lemma 2.14: Let $u: V^* \rightarrow \mathbb{R}$ be a harmonic function on the Vicsek set. If $v(x) = v(y) = 0$ and $y \in V_{m,x}$, then $v(\omega) = 0$ for all $\omega \in T_x$ where $T_x = \bigcup_{i_1, \dots, i_\alpha} \mathcal{S}_{i_1, i_2, \dots}(K)$.

Proof. It is clear $v(\omega) = 0$ for $\omega \in T_x \cap V^*$ since v is harmonic and depends on the values of its neighbours.

So we need to check the points in $T_x \setminus V^*$. Now since $T_x \cap V^*$ is dense in T_x , then for any $\omega \in T_x$ we have that there exists a sequence $x_n \in T_x \cap V^*$ such that $x_n \rightarrow \omega$. By the fact that v is continuous on K it is continuous on T_x , in which case $|v(x_n) - v(\omega)| \rightarrow 0$ as $n \rightarrow \infty$, but $v(x_n) = 0$ for all x_n , so $v(\omega) = 0$ as desired. □

Smaller corrections:

1. Page 4, line 2. "... for each $x \in F$..." should be "... for each $x \in K$..."
2. Page 6, Lemma 1.10, line 1. "... such that $\mu(F) \in (0, \infty)$..." should be "... such that $\mu(K) \in (0, \infty)$..."
3. Page 6, Lemma 1.10, line 1. "Let μ be a measure such..." should be "Let μ be an outer measure..."
4. Page 6, Lemma 1.11, line 3. "... intersects at most $(1 + 2a_2)^n a_1^n$..." should be "... intersects at most $(1 + 2a_2)^n a_1^{-n}$..."
5. Page 8, Line 9 from the bottom. " $x_{i_1, \dots, i_k} \in K \cap B$..." should be " $x_{i_1, i_2, \dots} \in K \cap B$..."

6. Page 14, Definition 2.5, line 1. "Let $\tilde{u} : V_{m+n} \rightarrow \mathbb{R}$ with $\tilde{u}|_{V_m} = u$ being a *harmonic function* if ..." should be "Let $\tilde{u} : V_{m+n} \rightarrow \mathbb{R}$ with $\tilde{u}|_{V_m} = u$. We call $\tilde{u}$ a *harmonic function* if ...".
7. Page 16, Definition 2.8, line 4. " $\text{dom } \mathcal{E} = \{u \in \ell(V_*) : \mathcal{E}(u) < \infty\}$ " should be " $\text{dom } \mathcal{E} = \{u \in \ell(V^*) : \mathcal{E}(u) < \infty\}$ ".
8. Page 17, line 16. "The constants 3^m and 5^m are not random, ..." should be "The constants 3^m and $(N_p)^{-1}$ from Δ_m are not random, ...".
9. Page 17, line 4 from the bottom. "... the values in $K \setminus V_m$ comes from ..." should be "... the values in $V^* \setminus V_m$ comes from ...".
10. Page 18, Lemma 2.14. "...".
11. Page 18, line 11 from the bottom. "... Then $F_m = K \setminus \left(\bigcup_{\alpha \in \Omega_m} \overline{B}_\alpha\right)$." should be "... Then $F_m = K \setminus \left(\bigcup_{\alpha \in \Omega_m} \overline{T}_\alpha\right)$."
12. Page 25, Definition 2.21, line 2. "... the *discrete fractional Laplacian* $(-\Delta_m)^s : L^2(V_m, \mu_m) \rightarrow L^2(V_m, \mu_m)$..." should be "... the *discrete fractional Laplacian* $(-\Delta_m)^{-s} : L^2(V_m, \mu_m) \rightarrow L^2(V_m, \mu_m)$...".
13. Page 29, line 13. "... generator of a strongly continuous Markov semigroup" should be "... generator of a strongly continuous self-adjoint contraction semigroup".
14. Page 31, line 5 & 16. "... $\frac{1}{\lambda_n^{-s}} \phi_n$." should be "... $\frac{1}{\lambda_n^s} \phi_n$."
15. Page 32, line 10 from the bottom. "... $\langle f, (-\Delta)^s g \rangle$." should be "... $\langle f, (-\Delta)^{-s} g \rangle$."
16. Page 33, line 1 from the bottom. " $|\phi_j| \leq C e^{\lambda_j/4}$ " should be " $|\phi_j(x)| \leq C e^{\lambda_j/4}$ ".
17. Page 36, line 6 from the bottom. "... $R 2^{-j-1} \leq d(x, y) \leq R 2^{-j-1} \Rightarrow 2^{-j-1} \leq R^{-1} \cdot d(x, y) \leq 2^{-j-1}$." should be "... $R 2^{-j-1} \leq d(x, y) \leq R 2^{-j-1}$."
18. Page 37, line 1. "Now choose k' such $u_{k'} \leq R$..." should be "Now choose k' such $u_{k'} \geq R$...".
19. Page 37, line 5. "We can likewise bound $G_s(x, y)$ by" should be "We can likewise bound $G_s(x, y)$ for $s < \frac{d_h}{d_w}$ by".
20. Page 41, line 5 from the bottom. "and ϕ is positive semidefinite" should be "and Φ is positive semidefinite"
21. Page 49, line 8. "By Lemma 4.1 and Corollary 2.26 we find that ..." should be "By Lemma 4.1 and Remark 2.28 we find that ..."
22. Page 50, line 6. "... the characteristic functional, e.i. that ..." should be "... the characteristic functional, i.e. that ...".
23. Page 53, line 6 from the bottom. "... so $-\Delta_m(M_{\text{eig}})_{i,j} = (v_{\text{eig}})_i \cdot (M_{\text{eig}})_{i,j} = \phi_i(x_j)$." should be "... $-\Delta_m(M_{\text{eig}})_{i,j} = (v_{\text{eig}})_i \cdot (M_{\text{eig}})_{i,j} = \lambda_i^m \phi_i(x_j)$."
24. Page 58, line 4. "... what values the simulation converges to." should be "... what values the simulation diverges to."